

7.2 EFFICIENCY & SOCIAL SURPLUS

Multiple Choice Questions

1985/AL/I/22

Consumer surplus is the

- A. excess quantity of a good which the consumer cannot consume.
- B. quantity of a good which the consumer chooses not to consume.
- C. extra satisfaction which the consumer enjoys in the purchase of a good.
- D. extra amount the consumer is willing to pay over and above what he actually pays.

1986/AL/I/15

Consumer surplus is

- A. the amount left over after a consumer makes a purchase.
- B. the quantity supplied to a consumer exceeding the quantity he prefers.
- C. the difference between what a consumer is willing to pay over and above what he actually pays.
- D. the area underneath the consumer's demand curve.

1990/AL/I/02

'Consumer equilibrium' means

- A. to each individual, the marginal benefit of a good is equal to its price.
- B. to each individual, the marginal benefit of every good is equal.
- C. to each individual, his total expenditure is equal to his total income.
- D. the marginal benefit of a good is the same for all consumers in the market.

1991/AL/I/08

Which of the following statements about 'benefit' is true?

- A. When a good has a benefit, it must have a market value.
- B. A good has a high market value if it has a high benefit.
- C. The total benefit of a good is the intrinsic value of that good.
- D. The marginal benefit of a good to a person depends on the quantity of that good which he already possesses.

1995/AL/I/03

Currently, with no entrance fee, the flow of visitors to the Hong Kong Park does not reach its full capacity. If the government now charges an entrance fee,

- A. the consumers' surplus will rise.
- B. the total benefit will decline.
- C. the marginal benefit will fall.
- D. the cost of maintenance will rise.

1995/AL/I/06

If floods in China destroyed half the output of rice, the total benefit of rice will

- A. rise but the total market value may rise or fall.
- B. fall but the total market value may rise or fall.
- C. fall but the total market value will rise.
- D. rise but the total market value will fall.

1997/AL/I/07

A free good

- A. is provided by the government free of charge.
- B. has no benefit and is not preferred.
- C. is a good the supply of which is so abundant that more of it is not preferred.
- D. is a gift of nature.

1998/AL/I/12

Due to exceptionally good weather, the supply of wheat has increased. The total benefit of wheat will _____ and the total market value _____.

- A. decrease will decrease
- B. increase will increase
- C. decrease may decrease or increase
- D. increase may decrease or increase

1998/AL/I/24

Mary spent \$500 to enter a lucky draw and won an airline ticket with a market price of \$3 000. The ticket was neither transferable nor refundable. Which of the following is correct?

- A. It is costless for Mary to use the ticket.
- B. (out of syllabus)
- C. The ticket must worth more than \$500 to Mary.
- D. None of the above.

2000/AL/I/10

A free good is a good that yields zero

- A. marginal benefit.
- B. average benefit.
- C. total benefit.
- D. consumer surplus.

2001/AL/I/19

A free good

- A. is less preferable than an economic good.
- B. has zero marginal benefit.
- C. is a good that has no demand.
- D. is a good provided by a generous person free of charge.

2002/AL/I/10

Market price

- A. is a criterion which determines who wins and who loses.
- B. is the maximum amount a consumer is willing to pay at the margin.
- C. may not be necessary in directing the use of scarce resources.
- D. All of the above.

2003/AL/I/11

In production, resource allocation is said to be efficient when

- A. the marginal cost of production equals the marginal benefit.
- B. the marginal benefit equals the product price.
- C. the marginal cost of production equals the marginal revenue.
- D. the marginal cost of production equals the average cost of production.

2006/AL/I/02

The marginal benefit of an individual on a good

- A. is observable.
- B. is asserted to be decreasing.
- C. always begins with a positive value.
- D. has to be expressed in terms of money.

2006/AL/I/23

Satisfaction of the efficiency condition implies that

- A. (out of syllabus)
- B. there is no scarcity of resources.
- C. the value of all resources is maximized.
- D. all costs of transactions are zero.

2008/AL/I/05

Consumer surplus is

- A. what a consumer possesses that is more than he can consume.
- B. what a consumer has purchased that is more than he intended.
- C. that part of benefit of a good exceeding the expenditure.
- D. that part of income a consumer donated to charity.

2010/AL/I/07

The market price of a good

- A. reflects its marginal benefit.
- B. is determined by its total benefit.
- C. is a competitive criterion that leads to inefficiency.
- D. will not exist in the absence of money.

2012/AL/I/01

A free good

- A. is provided by the government free of charge.
- B. has zero marginal benefit.
- C. is less preferable than an economic good.
- D. has no demand.

2012/AL/I/05

Susan says, "I won't eat any more pineapple unless you pay me!" Which of the following statements is correct?

- A. The demand curve of Susan cannot be drawn.
- B. The marginal benefit of pineapple to Susan is zero.
- C. (out of syllabus)
- D. Susan's demand for pineapple is perfectly inelastic.

2012/AL/I/07

If floods in Thailand have destroyed more than half of her crops, the total market value of the crops

- A. may rise or fall but the total benefit will rise.
- B. may rise or fall but the total benefit will fall.
- C. will rise but the total benefit will fall.
- D. will fall but the total benefit will rise.

2012/AL/I/16

A video-rental club charges \$20 per movie, or consumers can purchase a pre-paid coupon at a charge of \$100 for 10 movies. Which of the following statements is correct?

- A. The consumer surplus of a coupon purchaser is larger.
- B. The consumer surplus of a non-coupon purchaser is larger.
- C. The consumer surplus is roughly the same either way.
- D. There is not enough information to tell which way yields a larger consumer surplus.

PP/DSE/I/12

The following table shows the total cost schedule of a price-taking firm which has a fixed cost equal to \$20.

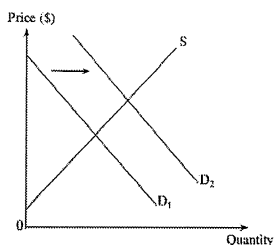
Quantity (units)	Total cost (\$)
1	40
2	65
3	95
4	130
5	170

Suppose the market price is \$35. At the profit-maximizing output level, the firm's profit and total producer surplus are _____ and _____ respectively.

- A. \$10 \$10
- B. \$10 \$30
- C. \$30 \$10
- D. \$30 \$30

PP/DSE/I/14

The following diagram shows the demand and supply curves of good X in a perfectly competitive market.



There is a shift in the demand curve for good X from D_1 to D_2 . As a result,

- A. firms enjoy more producer surplus.
- B. the marginal cost of producing the last unit decreases.
- C. the supply of good X increases.
- D. consumers enjoy less consumer surplus

PP/DSE/I/17

Economic efficiency is attained when

- (1) Total benefits to consumers are maximized.
 - (2) All potential gains are fully captured.
 - (3) Marginal social benefits are equal to marginal social costs.
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

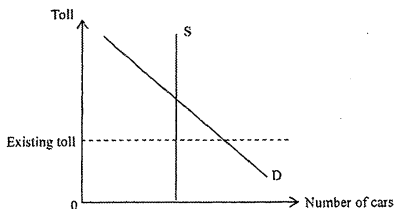
2012/DSE/I/14

On 10th July 2011, the urban taxi fare for the first 2 km was raised from \$18 to \$20. Which of the following statements about the urban taxi service are correct?

- (1) The demand for taxi service dropped.
 - (2) The price of long-distance taxi trips relative to short-distance taxi trips dropped.
 - (3) The consumer surplus for the taxi service dropped.
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2014/DSE/I/15

Refer to the following supply-demand diagram for the Cross-Harbour Tunnel.



The government proposed to raise the toll for the Cross-Harbour Tunnel so as to lessen the traffic congestion during rush hours. Suppose the maximum vehicular volume of the tunnel is fixed. Which of the following statements are correct?

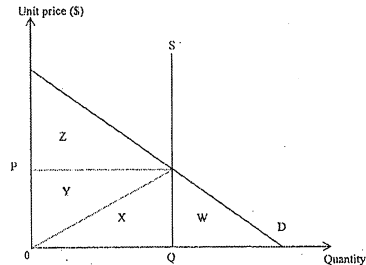
- (1) The demand for the service of the Cross-Harbour Tunnel will decrease.
 - (2) The total revenue of the Cross-Harbour Tunnel will increase.
 - (3) The deadweight loss caused by traffic congestion will decrease.
 - (4) The opportunity cost of the drivers using the tunnel will increase.
- A. (1) and (3) only
B. (1) and (4) only
C. (2) and (3) only
D. (2) and (4) only

2015/DSE/I/22

Refer to the following diagram.

The maximum total social surplus is equal to _____.

- A. Area Z.
- B. Area (Y+Z)
- C. Area (W+X+Y)
- D. Area (X+Y+Z)



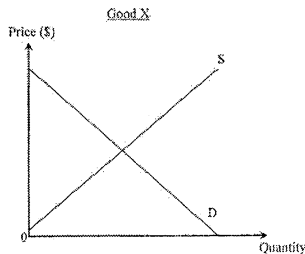
2016/DSE/I/21

In which of the following situations is efficiency attained?

- A. Total social benefit is equal to total social cost.
- B. Producer surplus is equal to consumer surplus.
- C. The difference between marginal social benefit and marginal social cost is zero.
- D. The consumer surplus of all consumers in the market is maximised.

2016/DSE/I/22

The following diagram shows the supply and demand curves of Good X.



Good X and Good Y are in competitive supply. A rise in the demand for Good Y will lead to in the consumer surplus and in the producer surplus in the market of Good X.

(Assume the supply and demand curves of Good X shift in a parallel manner.)

- A. an increase an increase
- B. an increase a decrease
- C. a decrease an increase
- D. a decrease a decrease

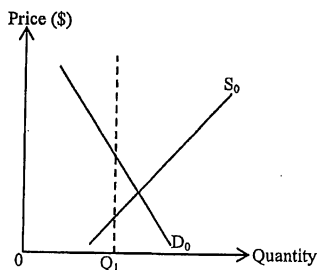
2017/DSE/I/12

A famous artist, Tsz-Wah, held a show. Many people complained that they could not buy a ticket. Ticket touts resold the tickets on the Internet and some buyers paid a price higher than HK\$10 000 for a ticket. The above case illustrates the _____ function of price because _____.

- A. rationing the tickets are rationed to the highest-valued buyers
- B. rationing demand will be reduced by a higher price
- C. allocative more profits can be earned with a higher price
- D. allocative more resources should be allocated to the producer to raise the supply

2019/DSE/I/21

Refer to the following supply-demand diagram of Good X.



If the government sets a quota at Q_1 , then _____ will decrease.

- (1) total social surplus
- (2) total expenditure on Good X
- (3) consumer surplus
- (4) producer surplus

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (4) only
- D. (3) and (4) only

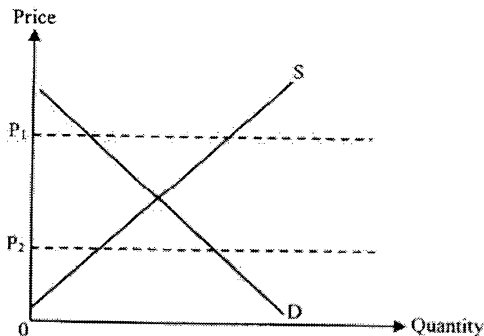
2019/DSE/I/22

Suppose there is a bad harvest of flowers due to global warming. The marginal benefit of flowers will _____ and the total exchange value _____.

- A. decrease.....will decrease
- B. increase.....will increase
- C. decrease.....may decrease or increase
- D. increase.....may decrease or increase

2021/DSE/I/17

Refer to the following demand-supply diagram of Good Y.



If the government lowers the price ceiling of Good Y from P_1 to P_2 ,

- A. the producer surplus will decrease.
- B. the consumer surplus will increase.
- C. the total social surplus will remain unchanged.
- D. the excess supply of Good Y will be eliminated.

2021/DSE/I/19

Since most schools deliver their lessons online, more students need to buy tablet computers. The marginal cost of the last unit of tablet computers will _____ and the producer surplus will _____.

- A. decrease ... increase
- B. decrease ... decrease
- C. increase ... increase
- D. increase ... decrease

2021/DSE/I/20

Initially a country imposes an effective quota on coffee bean production.

If now more people like to drink coffee, which of the following statements about coffee bean market are correct?

- (1) The total expenditure will increase.
 - (2) The deadweight loss will increase.
 - (3) The consumer surplus will increase.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

2022/DSE/I/10

In 2021, Trial & Error (試當真), a Hong Kong YouTube channel, held its first live show. The following was the pricing arrangement of its tickets.

Ticket price adjusts daily at 10 am.

Date	Ticket price
9 August	\$10 000
10 August	\$5 000
11 August	\$2 000
...	...
31 August	\$20
1 September onwards	\$10

The above case illustrates the _____ function of price because _____.

- A. allocative.....more resources can be allocated to the production of the show
- B. allocative.....the arrangement can guarantee a sell-out of the tickets
- C. rationing.....the tickets are sold to the highest-valued buyers
- D. rationing.....the tickets are distributed on a first-come, first-served basis

2022/DSE/I/20

The following table shows the marginal benefit schedule of a consumer for avocados.

Quantity (units)	1	2	3	4	5	6
Marginal benefit (\$)	8	7	6	5	4	3

When the price of avocados increases from \$5 to \$6, the consumer surplus will _____.

- A. increase by \$1
- B. increase by \$3
- C. decrease by \$1
- D. decrease by \$3

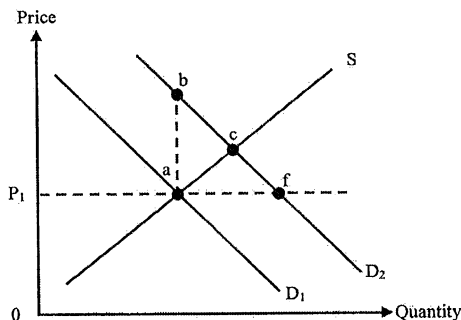
2023/DSE/I/20

The provision of a per-unit production subsidy on a good will lead to _____.

- A. a fall in the producer surplus of the good
- B. a rise in consumers' total expenditure on the good
- C. a rise in the total social surplus of the good
- D. a rise in the total revenue including the subsidy of the good

2023/DSE/I/21

The following diagram shows the supply and demand curves of Good Z.



Suppose a price ceiling is set at P_1 . When the demand for Good Z increases from D_1 to D_2 , there is _____ and the deadweight loss is _____.

- A. over-production.....area abc
- B. over-productionarea acf
- C. under-production.....area abc
- D. under-production area acf

2023/DSE/I/22

Plastic straws are a major source of plastic waste. Most plastic straws would break into smaller particles, releasing chemicals which threaten the environment.

Based on the above information, which of the following statements is correct?

- A. The use of plastic straws results in negative externality, and the quantity of plastic straws used is below the efficient level.
- B. To maximise total social surplus, governments around the world should completely ban the use of plastic straws.
- C. Charging a lump-sum tax on the businesses that provide plastic straws to customers must restore efficiency in the market of plastic straws.
- D. Granting a per-unit subsidy to the producers of environmentally-friendly paper straws tends to reduce the deadweight loss in the market of plastic straws.

2024/DSE/I/19

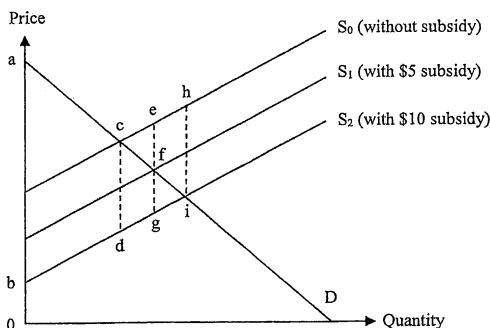
Which of the following statements about overproduction is/are correct?

- (1) Overproduction of a good occurs when there is an excess supply of the good.
- (2) Overproduction does not occur whenever the market is at equilibrium.
- (3) When overproduction occurs, the marginal social cost is greater than the marginal social benefit at the last unit.

- A. (2) only
- B. (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

2024/DSE/I/20

The following diagram shows the supply and demand curves of a good.

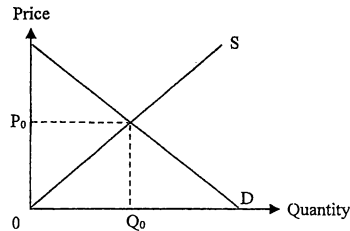


Suppose the government increases the per-unit subsidy from \$5 to \$10. Which of the following statements is correct?

- A. The new total social surplus is area aib.
- B. The new total social surplus is area acdb.
- C. The increase in deadweight loss is area ehif.
- D. The increase in deadweight loss is area cid,

2024/DSE/I/21

The following is the demand-supply diagram of Good X. Initially the price of Good X is P_0 .



Which of the following situations may reduce the consumer surplus and create a deadweight loss in the market of Good X at the same time?

- (1) a decrease *in* the demand for Good X
 - (2) the imposition of an effective price ceiling on Good X
 - (3) the imposition of an effective price floor on Good X
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

2024/DSE/I/22

In recent years, more and more people play videos loudly with their smartphones on public transport, disturbing other passengers.

Based on the above information, which of the following statements are correct?

- (1) Total social surplus may decrease if playing videos loudly with smartphones on public transport is banned by the government.
 - (2) Economic efficiency may improve if suppliers of public transportation services designate quiet zones in their vehicles where passengers are not allowed to play videos loudly with their smartphones.
 - (3) The divergence between private and social costs may decrease if people playing videos loudly with their smartphones on public transport are fined.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

2025/DSE/I/12

ABC Company is a food delivery service provider. It offers the following two payment options to customers.

Option 1: to pay a delivery fee of \$30 per order

Option 2: to pay a lump-sum membership fee of \$200 per month and do not need to pay any delivery fee per order

Based on the above information, after a customer paying the lump-sum membership fee, _____.

- (1) he/she tends to use the food delivery services of ABC Company more frequently
 - (2) he/she tends to order higher-priced food when using the food delivery services of ABC Company
 - (3) the marginal benefit of the last unit of food delivery services of ABC Company consumed by him/her tends to become lower
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

2025/DSE/I/18

Due to poor harvests of oranges, many juice producers use other fruits such as mandarins as an alternative ingredient to produce juice.

Based on the above information, which of the following is correct?

- A. There is a deadweight loss in the market of oranges.
- B. The total revenue of orange sellers may increase.
- C. The total social surplus in the market of mandarins decreases.
- D. The total exchange value of mandarins may decrease.

2025/DSE/I/21

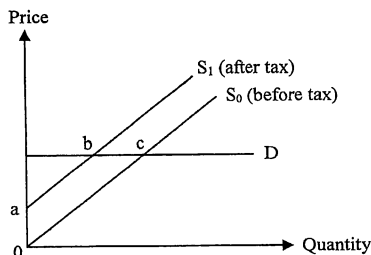
A developer plans to construct a tall building. The building will block the sea view enjoyed by some households living in another building.

In the above situation, efficiency may be attained if _____.

- (1) the government restricts the height of the new building
 - (2) the developer pays a compensation to each affected household for blocking the sea view
 - (3) the affected households pay the developer to request it to reduce the height of the new building
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

2025/DSE/I/22

The following diagram shows the supply and demand curves of a good. Suppose the government imposes a per-unit tax on the production of the good.



Which of the following statements about the imposition of the per-unit tax are correct?

- (1) The consumers bear no tax burden.
- (2) The decrease in total social surplus is area $0abc$.
- (3) There is deadweight loss after the imposition of the tax.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

2025/DSE/I/43

A small open economy has imposed an effective quota on imported red wine. If the world price of red wine decreases, ____.

- A. the volume of red wine imported by the economy will increase
- B. the quantity of red wine domestically produced in the economy will increase
- C. the quantity of red wine consumed in the economy will remain unchanged
- D. the price of red wine sold in the economy will decrease

Short & Structured Questions

1988/AL/I/2

'The maximum one is willing to pay is marginal benefit.' Why is the word maximum associated with marginal benefit?

1989/AL/I/1

What is price?

1992/AL/I/6

In the State of Washington, one who wishes to hunt for mountain goats must obtain a special goat permit. Each year approximately 8 000 people apply for goat permits. A draw is held to select 1 000 'winners'. Each winner may then purchase a goat permit for US\$20.00, entitling him to kill a maximum of one goat over a one-year period. Permits are not legally transferable, nor can they be resold, in any form. Each year about 250 permit holders succeed in killing a goat.

If the goat permits were allowed to be resold in a free market, would the new price be higher or lower? And would more goats be killed? Explain.

1994/AL/I/8

Compact discs of popular songs are now being reproduced on a massive scale in China. This reproduction is illegal, and the reproduced discs are sold in the market at significantly lower prices than the original discs.

From the standpoint of maximizing social gains, should the illegal practice of reproducing discs be firmly suppressed so that only the higher-priced original discs can be found on the market? Explain.

1997/AL/I/8(a)

University education in Hong Kong is heavily subsidized by the government. Under this system, admission to university as well as the selections of university and field of study are based on examination results. Imagine that there is an alternative system in which students in which students are admitted according to their willingness and ability to pay the highest price (tuition).

Of the numerous differences between these two systems, analyse only the following: How hard students will study in a university, when each and every university sets a firm standard requirement for graduation and the award of a degree.

2002/AL/I/06(b)

The Western Harbour Tunnel cost \$7 billion to build. After several years of operation the traffic flow remains far less than anticipated. Since the cost of serving an extra car is zero, in order to maximize wealth, the tunnel company simply charges a toll per car to maximize total revenue.

Some economists argue that if the cost of serving an extra car is zero, as it will be if the tunnel has no traffic congestion, then, to maximize social surplus, no toll should be charged. The counter argument to this is that if no toll is charged, no private enterprise will build the tunnel. Now the tunnel is already built, and the cost in the past is no longer an opportunity cost. Will society gain if the government purchases the tunnel and lets all cars use it free of charge?

2006/AL/I/5

Illustrate and explain why the efficiency condition is not satisfied in buffet dinners.

2006/AL/I/9

Whereas ladies from mainland China are coming to Hong Kong to buy the authentic handbags bearing prestigious brand names, ladies from Hong Kong are going to Shenzhen to buy fake handbags bearing the same brand names. The handbags look alike, but the authentic ones cost nearly 50 times more than the fake ones, even though they are only slightly more durable.

Does this imply that the ladies from China are richer than the Hong Kong ladies? Why would they pay such high prices for handbags which are expected to be only slightly more durable?

2009/AL/I/7(a)

In the United States, hunting of wildlife is restricted to certain seasons of the year. For residents of a given state, hunting licences for different species are required and each species bears a different licence fee. Out-of-state residents who want to hunt in the same state are required to pay higher licence fees.

Do you expect the out-of-state licence holders to be better hunters than the licence holders of the resident state? Explain.

PP/DSE/I/4

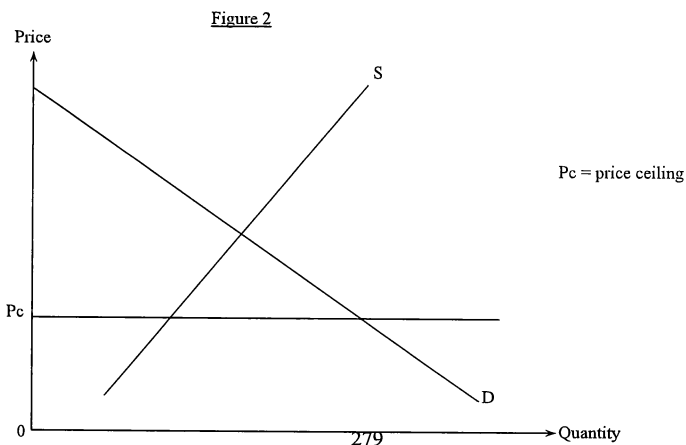
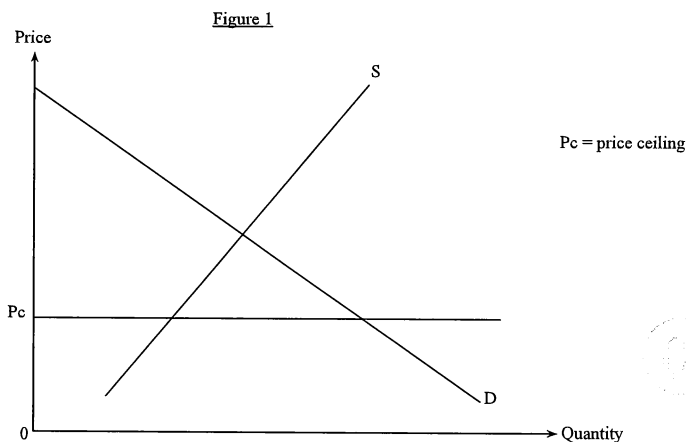
(a) What is "deadweight loss"? (2 marks)

(b)

A teacher said, "An effective price ceiling may increase the consumer surplus."
His student replied, "If this is the case, the price ceiling may enhance economic efficiency!"

(i) Do you agree with the teacher? Illustrate your answer in Figure 1. (3 marks)

(ii) Do you agree with the student? Explain your answer with the aid of Figure 2. (5 marks)



PP/DSE/I/12(b)(i)

In November 2010, the animated version of the Riverside Scene at the Qingming Festival (電子動態版清明上河圖) was exhibited in Hong Kong. 600 000 tickets were sold at \$10 each through the URBIX outlets and the Internet. Tickets were sold out within a short period of time. Many citizens complained that they could not purchase any tickets.

Some people paid \$200 to buy a ticket from touts (黃牛黨). State the function of price in the above case. (1 mark)

2015/DSE/I/4(a)

Read the following news extract.

The strong preference for small flats has pushed the prices of the flats under 400 square feet to a new high in the past few months. It is expected that more and more small flats will be available for sale in the coming future.

Explain **ONE** function of price which can be found in the above news extract. (2 marks)

2016/DSE/I/11

Medical service is one of the basic human needs.

Source A: a conversation outside a public clinic in Hong Kong

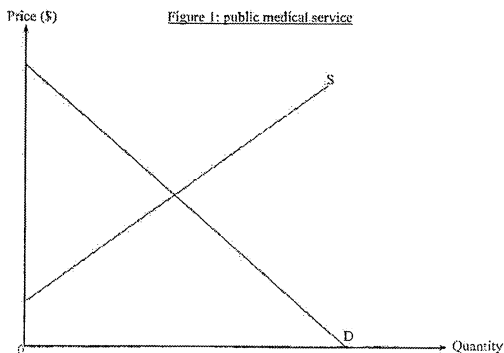
Son: Dad, I feel very ill.

Father: Oh no! There is a long queue waiting.

Son: Why don't we go to a private doctor?

Father: The fee charged by public clinics is only \$45, much lower than that charged by private doctors.

- (b) With reference to Source A, explain with the aid of Figure 1 how the price set by public clinics may lead to economic inefficiency in the provision of public medical service. (6 marks)
(Assume the supply curve in Figure 1 fully reflects the marginal social cost of public medical service.)



Source B: an extract from the Hospital Authority webpage

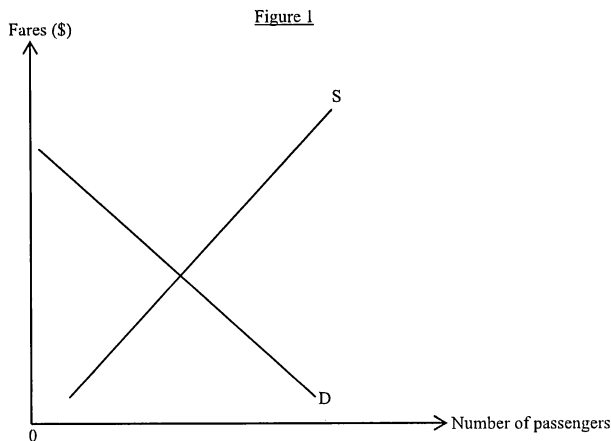
The Hospital Authority is committed to providing community-based primary care services. Our target patients include the elderly, low-income individuals, and patients with chronic diseases.

- (d) Many people argue that public clinics should raise the fee to reduce the waiting time. Evaluate this argument in terms of economic efficiency. (2 marks)

2019/DSE/II/11

The Guangzhou-Shenzhen-Hong Kong Express Rail Link (High Speed Rail) was launched in September 2018.

- (a) Explain the effect of the launch of High Speed Rail on labour mobility of Hong Kong. (3 marks)
- (b) During non-holidays, there are a lot of empty seats on some trains of the High Speed Rail. With the aid of Figure 1, explain why the above situation may be inefficient. (4 marks)



2023/DSE/II/5a-b

Refer to the following conversation.

Teacher: If the marginal cost for producing each unit of a good increases, the total social surplus will decrease.

Student: Following your argument, the increase in marginal cost for each unit will lead to economic inefficiency.

- (a) Illustrate the teacher's statement with a diagram. (2 marks)
- (b) Explain whether you agree with the student's response. (2 marks)

2023/DSE/II/11b

In Hong Kong, fans of popular singers often complain that they are unable to buy concert tickets of their idols at the official prices.

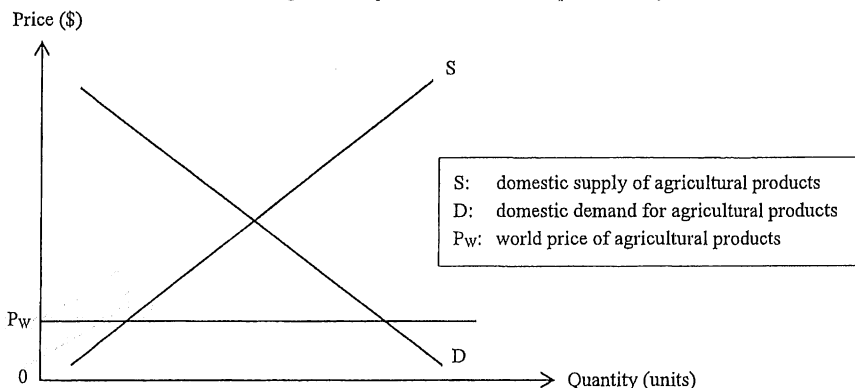
Some fans who are unable to purchase concert tickets through official channels would choose to buy tickets from touts (黄牛黨) at prices substantially higher than the official prices.

- (b) Which function of price is illustrated in the above situation? Explain your answer. (2 marks)

2025/DSE/II/8

A small open economy imports agricultural products from other countries although agricultural products are also produced by domestic farmers in the economy. To protect the domestic farmers, the government of the economy imposes a per-unit tariff on imported agricultural products. The economy remains to be an importer of agricultural products after the imposition of the tariff.

Figure 1: Market of agricultural products in the small open economy



(a) In Figure 1, indicate the following after the imposition of the tariff:

- the quantity of domestic production as Q_1 ,
- the import volume as Q_M , and
- the tariff revenue.

(4 marks)

(b) Explain the effect of the imposition of the tariff on the economy's trade balance.
(Hint: the tariff revenue is not included in the calculation of trade balance.)

(3 marks)